

Orbit–Scale CRT Compression and Uniform Gram Obstructions: Product–Slope Character Spectrum, Three–Modulus Alias Reduction, and a Coefficient–Specific Gate

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Abstract

The preceding paper reduces a distinguished zero–frequency shell to an orbit–sliced row energy

$$V_L + V_R \ll X^{o(1)} Q^3 / J,$$

whose proved diagonal is only $O(X^{o(1)} Q^2 J)$. Here we decide how far three natural routes can reach and identify the first coefficient–sensitive spectral quantity for the TPC-34 orbit–sliced product–slope interface.

First, for the abstract equal–row deletion model

$$A = \mathbf{11}^* - I_n, \quad M_j = AD_j,$$

with $t < n$ unit–phase observations, the off–diagonal Gram matrix is exactly

$$O = (n - 2)(R^* R - tI_n).$$

Consequently

$$\|O\|_{\text{op}} \geq (n - 2) \max\{t, n - t\}, \quad \|O\|_{\text{HS}}^2 \geq (n - 2)^2 nt(n - t),$$

with equality at the Fourier tight–frame scale. A block version survives same–source deletion. Thus cardinality, unit phases, and these deletions alone cannot certify the much smaller uniform operator or Hilbert–Schmidt gates proposed previously. This is an abstract information obstruction, not a lower bound for the physical masked Gram matrix.

Second, we prove that the residue–multiplicity implementation of one–prime projective Plancherel saturates at a normalized rms factor $\min\{J_0, S_0\}^{-1/2}$, with balanced residue sets attaining this scale up to constants. We then diagonalize a complete, separable coherent product–slope sum. For centered $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$, its exact singular spectrum is indexed by multiplicative characters. Projective Plancherel controls the average character energy, whereas a coherent weight sees its maximum; the possible average–to–maximum loss is exactly $q - 1$. This selects a coefficient–specific finite–model interface rather than a uniform norm gate. A controlled decomposition of the actual physical mask into such complete separable layers is not proved here.

Third, simultaneous orbit–scale congruences have an exact alias ladder. For r pairwise coprime moduli $q_i \asymp J$ and an affine determinant F of size $O(X)$, the number of possible aliases is $O(1 + X/J^r)$. At

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)},$$

three moduli leave $X/J^3 = X^{1/400+o(1)} = Q/J^2$, exactly the allowance above the orbit diagonal, while four moduli are the first orbit-scale choice giving strict no-wrap. We prove, however, that CRT coordinates do not automatically tensorize: data supported on one coordinate reproduce the one-prime normalized rms exactly. The specified merge-then-complete strategy also obeys the bookkeeping inequality $(1 + X/P)(1 + P/J) \geq X/J \asymp Q$.

On the physical side, the same-ultra block of the orbit-sliced energy satisfies

$$\mathcal{V}_{L,u=} + \mathcal{V}_{R,u=} \ll X^{o(1)} Q^2 J (1 + Q/T) \ll X^{o(1)} Q^3 J/T.$$

This removes one orbit factor from the earlier two-time estimate and saves the full factor T , but it still misses the orbit gate by J^2/T . The outcome is a precise next target: a joint three-modulus theorem that suppresses low-coordinate CRT modes in the actual coefficient direction, or an equivalent bound for the bad multiplicative-character mass, together with the missing complete separable reassembly lemma. No such physical coefficient estimate or reassembly is proved here. We claim no affine Chowla theorem, parity breach, Hardy-Littlewood asymptotic, or twin-prime result.

Keywords: Möbius correlation; orbit energy; Gram matrix; projective Plancherel; multiplicative character; CRT aliasing; parity boundary.

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1 Introduction

The matched residual chain now has a sharply localized unresolved object. TPC-32 isolates a distinguished zero-frequency shell, TPC-33 converts it to two physical column energies, and TPC-34 performs the corresponding TT^* expansion and orbit slicing [8–10]. After closing the complete Gram-copy diagonal, the remaining sufficient estimate is

$$V_L + V_R \ll X^{o(1)} \frac{Q^3}{J}. \quad (1)$$

Its proved diagonal is $O(X^{o(1)} Q^2 J)$. At the endpoint

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (2)$$

the ratio between the allowance in (1) and that diagonal is only

$$\frac{Q^3/J}{Q^2 J} = \frac{Q}{J^2} = X^{1/400+o(1)}. \quad (3)$$

Thus a useful next step must distinguish a genuine near-diagonal mechanism from a norm inequality whose worst direction is irrelevant to the physical coefficients.

This paper audits three routes suggested at the end of TPC-34: uniform Gram norms, one-modulus projective Plancherel, and multi-modulus or integer incidence. The purpose is not to manufacture a stronger claim from a larger formalism. It is to prove exactly which pieces close, which abstract information is insufficient, and what signed quantity a future theorem must control.

1.1 Uniform Gram norms: an exact aspect-ratio calculation

The operator and Hilbert-Schmidt conditions in TPC-34 were stated only as convenient sufficient interfaces. We show that they cannot follow from the broad data used by a sign-blind majorant. In the equal-deletion phase model, let

$$A = \mathbf{1}\mathbf{1}^* - I_n, \quad M_j = AD_j, \quad 1 \leq j \leq t < n,$$

where the diagonal entries of every D_j have modulus one. If R is the $t \times n$ phase matrix, then the off-diagonal part of $\sum_j M_j^* M_j$ is not merely bounded by a frame expression: it is exactly

$$O = (n - 2)(R^* R - tI_n). \quad (4)$$

Rank and trace alone therefore give

$$\|O\|_{\text{op}} \geq (n - 2) \max\{t, n - t\}, \quad (5)$$

$$\|O\|_{\text{HS}}^2 \geq (n - 2)^2 nt(n - t). \quad (6)$$

The bounds are attained by a partial Fourier matrix. At $n \asymp Q, t \asymp J$, their scales are Q^2 and $Q^4 J$, whereas the previous sufficient gates ask for Q^2/J and Q^4/J^2 . A block calculation shows that deleting every pair from the same prime source does not change this conclusion at the abstract scale.

The quantifier is essential. These theorems concern a class retaining only cardinality, unit-phase, and deletion data. The physical matrix has additional gcd projectors, near-row deletion, smooth weights, Möbius zeros, and a distinguished coefficient vector. Entry deletion is not an order operation for a Hermitian off-diagonal matrix. We therefore do *not* prove that the physical operator is large or that its uniform gate is false. We prove that the listed coarse data cannot prove that it is small.

1.2 From projective averages to a multiplicative spectrum

TPC-33 proves an exact projective Plancherel identity over $\mathbb{F}_q$. Its incomplete integer implementation is controlled by maximum residue multiplicities. We normalize that right-hand side by the pointwise Cauchy scale and prove the universal lower bound

$$\mathcal{R}_q \geq \min\{J_0, S_0\}^{-1/2}. \quad (7)$$

Balanced nonzero-residue sets attain this scale up to constants. This is a saturation theorem for that specific residue-multiplicity certificate; it is not a lower bound for the actual correlation energy and not a no-go theorem for all projective methods.

The positive result is an exact diagonalization of a complete, separable coherent product-slope operator. If

$$R_s(a) = \sum_D f(D)g(s^{-1}(aD + h)), \quad (Tb)(j, s) = \sum_{\ell \neq 0} b(\ell)R_s(\ell j),$$

then multiplicative characters diagonalize T :

$$\sum_{j,s} |Tb(j, s)|^2 = \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \mathcal{E}_{\chi}, \quad \|T\|_{\text{op}}^2 = \max_{\chi} \mathcal{E}_{\chi}. \quad (8)$$

The projective second moment is the average of the same $\mathcal{E}_{\chi}$. Hence the passage from independent slope averaging to one coherent coefficient vector is precisely an average-versus-maximum problem in multiplicative character space. The factor $q - 1$ can be attained by a single-character finite model. At $q \asymp J$, it is exactly the energy factor that the one-modulus route fails to supply. Applying this identity to the physical orbit energy would additionally require a controlled decomposition of its j -dependent mask and incomplete ranges into complete separable layers. That reassembly lemma is not proved here.

1.3 The critical CRT ladder

For an integer affine determinant F with $|F| \leq B \asymp X$, simultaneous congruences modulo $q_1, \dots, q_r$ leave the exact aliases

$$F = k [q_1, \dots, q_r], \quad |k| \leq B/[q_1, \dots, q_r].$$

Taking pairwise coprime orbit-scale moduli $q_i \asymp J$ gives

$$\frac{X}{J^r} = X^{1-r(133/400)+o(1)}. \quad (9)$$

Three moduli are the first to reach the margin in (3); four are the first to make the ambient box strictly no-wrap. The numerical match

$$\frac{X}{J^3} = \frac{Q}{J^2} = X^{1/400+o(1)} \quad (10)$$

is exact at exponent level.

It is only a route ledger. We prove two reasons why separate one-modulus estimates cannot simply be multiplied. First, a function lifted from one CRT coordinate has exactly the same normalized projective rms at the product modulus as it had at that one prime. Second, the specified strategy of merging moduli to a single modulus P and then completing the J -long orbit incurs costs whose product obeys the bookkeeping inequality

$$(1 + X/P)(1 + P/J) \geq X/J \asymp Q. \quad (11)$$

A viable three-modulus theorem must therefore keep the coordinates joint and suppress their principal and proper-coordinate faces in the actual coefficient direction.

1.4 A physical gain and the remaining boundary

Opening both ultra increments inside the orbit-sliced energy, we bound the block in which the two copies use the same divisor $u > T$:

$$\mathcal{V}_{L,u} + \mathcal{V}_{R,u} \ll X^{o(1)} Q^2 J(1 + Q/T). \quad (12)$$

At the endpoint it is $O(X^{o(1)} Q^3 J/T)$. This is a true theorem for the physical mask and coefficient system. It is one factor J smaller than the analogous two-time block in TPC-34, and it saves the full T from the orbit-sliced absolute baseline. It nevertheless exceeds the required Q^3/J by J^2/T .

We also record the exact short-orbit incidence operator and an ambient integer-line Schur bound. They explain why congruence uniqueness is valuable but does not by itself control how many different divisor atoms can land at the same orbit time.

1.5 Claim boundary and organization

The conclusions are route diagnostics and exact finite spectral identities. They do not establish the coefficient-specific physical off-diagonal estimate. In particular, this paper proves no suppression of all bad multiplicative characters, no controlled complete separable decomposition of the physical mask, no joint three-modulus dispersion theorem, no affine Chowla estimate, no positivity statement, no Hardy-Littlewood prime-pair asymptotic, no infinitely many twin primes, and no breach of sieve parity.

Section 2 restates the physical interface. Section 3 proves the aspect-ratio theorems. Section 4 proves one-modulus saturation. Section 5 gives the exact character spectrum. Section 6 develops the CRT ladder and non-tensorization. Section 7 proves the physical same-ultra estimate. Section 8 treats short-orbit and integer incidence. The final two sections give the endpoint ledger, next gate, certificate, and exact claim boundary.

2 The inherited orbit-sliced gate

We use the physical notation and proved estimates of TPC-32–TPC-34 [8–10]. They are restated to separate theorems about the actual Möbius kernel from the abstract route diagnostics below.

2.1 Rows, orbit, and ultra increment

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and work on a dyadic packet with

$$Q = LD, \quad QJ \asymp X, \quad R \leq T < U_0 \asymp X. \quad (13)$$

The inherited source-compatible packet also satisfies, for a fixed $0 < \delta < 1/2$,

$$R = X^{1/2-\delta+o(1)}, \quad D \leq R^{1/2+o(1)}, \quad L > \max\{2R, 2|h|\}. \quad (14)$$

A row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad (15)$$

where $\ell_\alpha \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, and $(d_\alpha, H) = 1$. The scale separation in (14) makes the integer m_α determine the row label α uniquely. There are $O(Q)$ rows. The two physical coefficient systems obey

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll X^{o(1)} Q, \quad (16)$$

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (17)$$

The logarithm in the original L^2 bound is absorbed into $X^{o(1)}$.

For j in an interval of length $O(J)$, put

$$N_\alpha(j) = m_\alpha j + h \asymp X. \quad (18)$$

The inherited packet and its terminal horizon are chosen so that

$$T < \min_{\alpha, j} N_\alpha(j) \leq \max_{\alpha, j} N_\alpha(j) < U_0. \quad (19)$$

This is the terminal-range condition of TPC-34; in particular, the full target $N_\alpha(j)$ is one of the ultra divisors retained below. Every divisor $w \mid N_\alpha(j)$ on the physical support satisfies

$$(w, m_\alpha j H) = 1. \quad (20)$$

In particular $(w, j) = (w, h) = (w, m_\alpha) = 1$.

The actual bounded multiplier is

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > QX^{-\kappa_0}} \mathbf{1}_{(d_\alpha, d_\gamma) \leq X^{\kappa_0}} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J), \quad (21)$$

where the periodic and smooth factors are uniformly bounded. Define

$$a_R(u) = -\mu(u) \log u, \quad C_m(j) = \sum_{\substack{T < u \leq U_0 \\ u \mid mj+h}} a_R(u). \quad (22)$$

Divisor bounds give $|C_m(j)| \ll X^{o(1)}$, uniformly on the packet.

2.2 Orbit slicing and Gram decomposition

The left orbit slices and their energy are

$$Y_{\gamma,j}^L := \sum_{\alpha} \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad (23)$$

$$V_L := \sum_{\gamma,j} |Y_{\gamma,j}^L|^2. \quad (24)$$

The right objects exchange the two row systems:

$$Y_{\alpha,j}^R := \sum_{\gamma} \gamma_{\gamma}^{(2)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\gamma}}(j), \quad (25)$$

$$V_R := \sum_{\alpha,j} |Y_{\alpha,j}^R|^2. \quad (26)$$

TPC-34 proves that

$$V_L + V_R \ll X^{o(1)} \frac{Q^3}{J} \quad (27)$$

is sufficient for the inherited scalar shell estimate. It also proves the complete input–copy diagonal bound

$$V_{L,\Delta} + V_{R,\Delta} \ll X^{o(1)} Q^2 J. \quad (28)$$

For each j , let

$$(M_j^L)_{\gamma,\alpha} = \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad G_L^{\text{orb}} = \sum_j (M_j^L)^* M_j^L. \quad (29)$$

Write $G_L^{\text{orb}} = D_L^{\text{orb}} + O_L^{\text{orb}}$, with D the entrywise diagonal, and define the right objects symmetrically. Then

$$V_L = \langle G_L^{\text{orb}} \gamma^{(1)}, \gamma^{(1)} \rangle. \quad (30)$$

The previously displayed uniform sufficient condition is

$$\|O_L^{\text{orb}}\|_{\text{op}} + \|O_R^{\text{orb}}\|_{\text{op}} \ll X^{o(1)} \frac{Q^2}{J}, \quad (31)$$

and a Hilbert–Schmidt sufficient interface has squared scale

$$\|O_L^{\text{orb}}\|_{\text{HS}}^2 + \|O_R^{\text{orb}}\|_{\text{HS}}^2 \ll X^{o(1)} \frac{Q^4}{J^2}. \quad (32)$$

Neither estimate is inherited as a theorem.

The direct coefficient–specific sufficient condition already identified in TPC-34 is

$$|V_{L,\neq}| + |V_{R,\neq}| \ll X^{o(1)} \frac{Q^3}{J}. \quad (33)$$

It is stronger than logically necessary, because only the upper bound for their sum is needed, but it is much weaker than controlling every vector as in (31). The role of this paper is to show why the two convenient uniform routes saturate in broad abstract classes and to identify a complete finite-model spectrum that could feed into (33) after a separate physical reassembly lemma; we do not claim to discover the coefficient gate for the first time.

2.3 Endpoint scales

At the high- β packet,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}. \quad (34)$$

The underlying source and row scales are

$$L = X^{99979/210000+o(1)}, \quad D = X^{10049/52500+o(1)}. \quad (35)$$

In particular,

$$D = o(Q), \quad J = o(Q), \quad DJ = o(Q^2), \quad D \leq X/T \quad (36)$$

for sufficiently large X . The basic identities used repeatedly below are

$$QJ = X^{1+o(1)}, \quad \frac{Q}{J^2} = X^{1/400+o(1)}, \quad (37)$$

$$\frac{T}{J} = X^{107/2000+o(1)}, \quad \frac{J^2}{T} = X^{279/1000+o(1)}. \quad (38)$$

Every assertion involving these exponents is an endpoint accounting statement; it does not replace a uniform estimate on the full parameter range.

3 Aspect-ratio obstructions to uniform Gram gates

We first ask whether the uniform conditions (31)–(32) could follow from the most visible geometry: n input rows, t orbit times, bounded phases, and deletion of the equal row. The answer is no for that abstract information class. The calculation is exact and is a frame identity rather than a probabilistic counterexample. TPC-34 already gives one coherent constant-phase witness at the sign-blind scale [10]. The theorem below upgrades that witness to rank-trace lower bounds valid for every unit-phase equal-deletion frame and identifies the sharp Fourier case.

3.1 Equal-row deletion

Let $n \geq 3$, $1 \leq t < n$, and let

$$A = \mathbf{1}\mathbf{1}^* - I_n. \quad (39)$$

For $1 \leq j \leq t$, choose phases $c_{j\alpha} \in \mathbb{C}$ with $|c_{j\alpha}| = 1$, and put

$$D_j = \text{diag}(c_{j1}, \dots, c_{jn}), \quad M_j = AD_j. \quad (40)$$

Let $R = (c_{j\alpha})_{j,\alpha} \in \mathbb{C}^{t \times n}$, define

$$G = \sum_{j=1}^t M_j^* M_j, \quad (41)$$

and let $O = G - \text{diag } G$.

Theorem 3.1 (Exact equal-deletion Gram identity). *With the preceding notation,*

$$G = tI_n + (n-2)R^*R, \quad (42)$$

$$\text{diag } G = (n-1)tI_n, \quad (43)$$

$$\boxed{O = (n-2)(R^*R - tI_n)}. \quad (44)$$

Consequently

$$\boxed{\|O\|_{\text{op}} \geq (n-2) \max\{t, n-t\}}, \quad (45)$$

and

$$\boxed{\|O\|_{\text{HS}}^2 \geq (n-2)^2 nt(n-t)}. \quad (46)$$

Both lower bounds are equalities when R consists of any t rows of an $n \times n$ complex Fourier matrix.

Proof. Writing $J_n = \mathbf{1}\mathbf{1}^*$, one has

$$A^2 = (J_n - I_n)^2 = I_n + (n-2)J_n. \quad (47)$$

Moreover,

$$\sum_{j=1}^t D_j^* J_n D_j = R^* R.$$

This proves (42). Every diagonal entry of $R^* R$ is t , which gives (43) and (44).

Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $R^* R$. They are nonnegative,

$$\text{rank}(R^* R) \leq t, \quad \sum_i \lambda_i = \text{tr}(R^* R) = nt. \quad (48)$$

Because $t < n$, zero is an eigenvalue, so $\|R^* R - tI\|_{\text{op}} \geq t$. The largest eigenvalue is at least n , since at most t nonzero eigenvalues have sum nt . Hence the same norm is at least $n - t$, proving (45).

For the Hilbert–Schmidt norm, Cauchy–Schwarz on the nonzero eigenvalues gives

$$\sum_i \lambda_i^2 \geq \frac{(nt)^2}{t} = n^2 t.$$

Therefore

$$\begin{aligned} \|R^* R - tI\|_{\text{HS}}^2 &= \sum_i (\lambda_i - t)^2 \\ &= \sum_i \lambda_i^2 - 2t(nt) + nt^2 \\ &\geq nt(n-t), \end{aligned}$$

which proves (46).

For t Fourier rows, $RR^* = nI_t$. Thus the eigenvalues of $R^* R$ are n with multiplicity t and zero with multiplicity $n - t$, giving equality throughout. $\square$

Corollary 3.2 (Aspect-ratio gap). *If $n \asymp Q$, $t \asymp J$, and $J = o(Q)$, then every matrix in the model satisfies*

$$\|O\|_{\text{op}} \gg Q^2, \quad \|O\|_{\text{HS}}^2 \gg Q^4 J. \quad (49)$$

Relative to (31) and (32), these scales are larger by factors J and J^3 , respectively.

Proof. Since $n - t \asymp Q$, the operator lower bound is $\gg n(n - t) \asymp Q^2$. The Hilbert–Schmidt lower bound is

$$(n-2)^2 nt(n-t) \asymp n^4 t \asymp Q^4 J.$$

The ratios to Q^2/J and Q^4/J^2 are J and J^3 . $\square$

3.2 Same-source block deletion

The physical mask also deletes row pairs sharing the same prime source. The following block model shows that this deletion does not, by itself, remove the aspect–ratio effect.

Suppose $n = kr$ and partition the coordinates into k blocks of size r . Put

$$B = \bigoplus_{g=1}^k J_r, \quad A = J_n - B. \quad (50)$$

Thus $A_{\alpha\beta} = 0$ when the two coordinates lie in the same source block. Retain the phase matrices D_j and R above; let $R_g \in \mathbb{C}^{t \times r}$ be the columns of R in block g .

Theorem 3.3 (Exact same-source block formula). *Assume $n \geq 2r$. For $M_j = AD_j$, let $G = \sum_j M_j^* M_j$ and $O = G - \text{diag } G$. Then*

$$A^2 = (n - 2r)J_n + rB, \quad (51)$$

and

$$O = (n - 2r)(R^* R - tI_n) + r \bigoplus_{g=1}^k (R_g^* R_g - tI_r). \quad (52)$$

If $t < n$, then

$$\|O\|_{\text{op}} \geq (n - 2r)(n - t) - rt \quad (53)$$

whenever the right-hand side is positive.

Proof. The block relations

$$J_n^2 = nJ_n, \quad J_n B = B J_n = rJ_n, \quad B^2 = rB$$

give (51). Conjugating by D_j and summing gives

$$G = (n - 2r)R^* R + r \bigoplus_g R_g^* R_g.$$

Every diagonal entry equals $t(n - r)$, yielding (52).

Choose a unit eigenvector x for the largest eigenvalue λ of $R^* R$. As in (48), $\lambda \geq n$. Writing $x = \bigoplus_g x_g$, positivity gives

$$\begin{aligned} \langle O x, x \rangle &= (n - 2r)(\lambda - t) + r \sum_g (\|R_g x_g\|_2^2 - t\|x_g\|_2^2) \\ &\geq (n - 2r)(n - t) - rt. \end{aligned}$$

If this number is positive, it is bounded above by $\|O\|_{\text{op}}$. □

In the corresponding exponent-scale block model,

$$n = QX^{o(1)}, \quad r = DX^{o(1)}, \quad t = JX^{o(1)}.$$

The inherited source separation (14) and endpoint scales (35) imply $D = o(Q)$, $J = o(Q)$, and $DJ = o(Q^2)$. Hence the right side of (53) has exponent scale $Q^2 X^{o(1)}$. The same-source deletion alone therefore does not supply the factor J required by the uniform operator gate.

3.3 What the obstruction does and does not prove

Corollary 3.4 (Coarse-data method boundary). *No argument valid for every equal-deletion phase model (40), using only n, t , unit-modulus phase envelopes, and equal-row deletion, can certify the uniform scales in (31) or (32) when $n \asymp Q, t \asymp J \rightarrow \infty$. The same conclusion holds after same-source deletion for the block class above whenever the right side of (53) is $\asymp Q^2$.*

The corollary is a quantifier statement about an abstract majorant class. It does not say that the actual O_L^{orb} or O_R^{orb} has size Q^2 . The physical entries contain arithmetic zeros, an opened gcd projector, near-row deletion, smooth weights, and Möbius coefficients. Nor may one infer a physical lower bound by deleting entries from the abstract O : an off-diagonal Hermitian matrix has no entrywise monotonicity for its operator norm. The calculation instead forces the next proof to use some information absent from the broad phase model. The physical vector $\gamma^{(i)}$ in (30) is the most economical such information.

4 Saturation of the residue-multiplicity projective route

TPC-33 proves an exact determinant-fiber second moment over $\mathbb{F}_q$ and extends it to incomplete integer slope sets by grouping elements into residue classes. We now measure the strongest normalized factor that this specific multiplicity bound can certify. This should not be confused with a lower bound on the correlations being estimated.

4.1 Normalization

Let q be prime and let $\mathcal{J}, \mathcal{S}$ be finite integer sets, with no element divisible by q . Write

$$J_0 = |\mathcal{J}|, \quad S_0 = |\mathcal{S}|, \quad (54)$$

and define

$$\nu_{\mathcal{A}}(r) = \#\{a \in \mathcal{A} : a \equiv r \pmod{q}\}, \quad \nu_{\mathcal{A}}^* = \max_{r \in \mathbb{F}_q^\times} \nu_{\mathcal{A}}(r). \quad (55)$$

The incomplete projective estimate in TPC-33 has, after suppressing the common coefficient and function norms, the right-hand side

$$q\nu_{\mathcal{J}}^*\nu_{\mathcal{S}}^*.$$

The pointwise Cauchy bound summed over the $J_0 S_0$ slope pairs has scale $J_0 S_0$. We therefore define the squared certificate factor

$$\mathcal{R}_q^2(\mathcal{J}, \mathcal{S}) := \frac{q\nu_{\mathcal{J}}^*\nu_{\mathcal{S}}^*}{J_0 S_0}. \quad (56)$$

This is not the “rms loss relative to complete residue averaging” used in TPC-33. It is a different normalization: the ratio of that multiplicity right-hand side to the total pointwise Cauchy scale.

4.2 Universal lower bound and balanced-set sharpness

Theorem 4.1 (Incomplete multiplicity saturation). *For every prime q and every nonempty pair $\mathcal{J}, \mathcal{S}$ as above,*

$$\boxed{\mathcal{R}_q^2(\mathcal{J}, \mathcal{S}) \geq \frac{1}{\min\{J_0, S_0\}}, \quad \mathcal{R}_q(\mathcal{J}, \mathcal{S}) \geq \min\{J_0, S_0\}^{-1/2}.} \quad (57)$$

Conversely, if $M := \min\{J_0, S_0\} \geq 2$, there are a prime q and balanced nonzero-residue sets of those cardinalities for which

$$\boxed{\mathcal{R}_q^2(\mathcal{J}, \mathcal{S}) < \frac{4}{M}}. \quad (58)$$

Thus the exponent in (57) is sharp for this certificate.

Proof. By symmetry assume $J_0 \leq S_0$. Pigeonhole gives

$$\nu_{\mathcal{A}}^* \geq \max \left\{ 1, \frac{|\mathcal{A}|}{q-1} \right\}. \quad (59)$$

There are three ranges.

If $q \leq J_0$, apply the second term of (59) to both sets:

$$\mathcal{R}_q^2 \geq \frac{q}{(q-1)^2} \geq \frac{1}{q} \geq \frac{1}{J_0}.$$

If $J_0 < q \leq S_0$, use $\nu_{\mathcal{J}}^* \geq 1$ and $\nu_{\mathcal{S}}^* \geq S_0/(q-1)$:

$$\mathcal{R}_q^2 \geq \frac{q}{J_0(q-1)} \geq \frac{1}{J_0}.$$

Finally, if $q > S_0$, both multiplicities are at least one and

$$\mathcal{R}_q^2 \geq \frac{q}{J_0 S_0} > \frac{1}{J_0}.$$

This proves (57).

For sharpness, Bertrand's postulate supplies a prime $J_0 < q \leq 2J_0$. Choose $\mathcal{J}$ with one element in each of J_0 distinct nonzero residue classes, so $\nu_{\mathcal{J}}^* = 1$. Distribute the S_0 elements of $\mathcal{S}$ as evenly as possible among the $q-1$ nonzero classes. Then

$$\nu_{\mathcal{S}}^* \leq \frac{S_0}{q-1} + 1.$$

Integer representatives with these residue multiplicities plainly exist. Therefore

$$\mathcal{R}_q^2 \leq \frac{q}{J_0(q-1)} + \frac{q}{J_0 S_0} < \frac{2}{J_0} + \frac{2}{S_0} \leq \frac{4}{J_0},$$

which proves (58). □

Remark 4.2 (The continuous proxy). The algebraic envelope

$$\Phi(q) = \frac{1}{q} + \frac{1}{J_0} + \frac{1}{S_0} + \frac{q}{J_0 S_0}$$

is minimized at $q = \sqrt{J_0 S_0}$, with value $(J_0^{-1/2} + S_0^{-1/2})^2$. This is a useful continuous ledger and is checked by the certificate. It is not asserted to be an exact coefficient-one upper bound for an interval after its zero residue class has been removed.

Remark 4.3 (What is saturated). The theorem lower-bounds the best right-hand side factor available from maximum residue multiplicities. The actual projective correlation may be zero even when $\mathcal{R}_q$ is large. A method using cancellation between residue classes, weights adapted to the coefficient vector, more than one modulus jointly, or a direct integer large-sieve estimate is not covered by the theorem; those belong to the broader large-sieve framework [1, 3].

4.3 Endpoint consequence

In the orbit-slope application, $J_0 \asymp J$, while the other physical slope range may be shorter or longer. Since $\min\{J_0, S_0\} \leq J_0$, even the most optimistic amplitude factor certified by the one-modulus multiplicity right-hand side cannot improve on $J^{-1/2}$, or J^{-1} at energy level. Applied optimistically to the orbit-sliced absolute scale $Q^3 J$, it reaches only

$$Q^3 J \cdot J^{-1} = Q^3. \quad (60)$$

The desired energy is Q^3/J . With a common denominator 2000,

$$Q^3 J = X^{4670/2000+o(1)}, \quad Q^3 = X^{4005/2000+o(1)}, \quad \frac{Q^3}{J} = X^{3340/2000+o(1)}. \quad (61)$$

Thus this certificate misses exactly one further factor J at energy level. The next section identifies that factor as a possible average-to-maximum loss in multiplicative character space.

5 Exact multiplicative spectrum of the product-slope operator

The complete product-slope identity in TPC-33 sums the squared correlation separately over each source coefficient ℓ . In the physical row sum, the ℓ -coefficients are combined coherently before squaring. Multiplicative Fourier analysis makes the distinction exact.

5.1 Definitions and diagonalization

Let q be prime, put $G = \mathbb{F}_q^\times$, and fix $h \in G$. Let $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$ be centered:

$$\sum_{x \in \mathbb{F}_q} f(x) = \sum_{x \in \mathbb{F}_q} g(x) = 0. \quad (62)$$

For $a, s \in G$, define

$$R_s(a) = \sum_{D \in \mathbb{F}_q} f(D) g(s^{-1}(aD + h)). \quad (63)$$

For $b : G \rightarrow \mathbb{C}$, the product-slope operator is

$$(Tb)(j, s) = \sum_{\ell \in G} b(\ell) R_s(\ell j), \quad j, s \in G. \quad (64)$$

For a multiplicative character χ of G , use the unnormalized transform

$$\widehat{b}(\chi) = \sum_{x \in G} b(x) \overline{\chi(x)}, \quad \widehat{R}_s(\chi) = \sum_{a \in G} R_s(a) \overline{\chi(a)}, \quad (65)$$

and put

$$\mathcal{E}_\chi = \sum_{s \in G} |\widehat{R}_s(\chi)|^2. \quad (66)$$

Theorem 5.1 (Product-slope singular spectrum). *With the preceding conventions,*

$$\boxed{\sum_{j, s \in G} |(Tb)(j, s)|^2 = \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\chi)|^2 \mathcal{E}_\chi.} \quad (67)$$

Consequently

$$\boxed{\|T\|_{\text{op}}^2 = \max_{\chi} \mathcal{E}_\chi.} \quad (68)$$

For every nonprincipal character χ , extended by $\chi(0) = 0$,

$$\mathcal{E}_\chi = \left| \sum_{D \neq 0} f(D) \chi(D) \right|^2 \left(q \|g\|_2^2 - \left| \sum_{U \neq 0} g(U) \overline{\chi(U)} \right|^2 \right). \quad (69)$$

For the principal character $\mathbf{1}$,

$$\mathcal{E}_\mathbf{1} = q^2 |f(0)|^2 (\|g\|_2^2 - |g(0)|^2). \quad (70)$$

Here $\|g\|_2^2 = \sum_{U \in \mathbb{F}_q} |g(U)|^2$.

Proof. For fixed s , take the multiplicative transform in j . The substitution $a = \ell j$ gives

$$\begin{aligned} \sum_{j \in G} (Tb)(j, s) \overline{\chi(j)} &= \sum_{\ell \in G} b(\ell) \chi(\ell) \sum_{a \in G} R_s(a) \overline{\chi(a)} \\ &= \widehat{b}(\bar{\chi}) \widehat{R}_s(\chi). \end{aligned}$$

Multiplicative Parseval, followed by summation over s , proves (67). Parseval also gives

$$\sum_{\ell \in G} |b(\ell)|^2 = \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\chi)|^2.$$

The quotient of the two diagonal forms is maximized by supporting the transform of b on a character of maximal $\mathcal{E}_\chi$, proving (68).

Now let $\chi \neq \mathbf{1}$. The $D = 0$ part of $\widehat{R}_s(\chi)$ vanishes because $\sum_{a \in G} \overline{\chi(a)} = 0$. For $D \neq 0$, make the change of variables $U = s^{-1}(aD + h)$. Extending χ by zero permits the omitted value $a = 0$ to be restored, and gives

$$\widehat{R}_s(\chi) = \left(\sum_{D \neq 0} f(D) \chi(D) \right) \left(\sum_{U \in \mathbb{F}_q} g(U) \overline{\chi(sU - h)} \right). \quad (71)$$

As s ranges over G , $t = h/s$ also ranges over G , and the modulus of the second factor equals that of $\sum_U g(U) \overline{\chi(U - t)}$. If $t = 0$ is temporarily included, additive Fourier Parseval and the standard nontrivial-character Gauss-sum modulus $\sqrt{q}$ give

$$\sum_{t \in \mathbb{F}_q} \left| \sum_U g(U) \overline{\chi(U - t)} \right|^2 = q \|g\|_2^2. \quad (72)$$

The zero additive frequency contributes nothing because g is centered. Removing $t = 0$ subtracts $|\sum_{U \neq 0} g(U) \overline{\chi(U)}|^2$, proving (69).

For $\chi = \mathbf{1}$, sum (63) over $a \neq 0$. When $D \neq 0$, the affine variable $s^{-1}(aD + h)$ runs over $\mathbb{F}_q \setminus \{h/s\}$, so centeredness of g gives $-g(h/s)$. When $D = 0$, the contribution is $(q-1)f(0)g(h/s)$. Since centeredness of f gives $\sum_{D \neq 0} f(D) = -f(0)$,

$$\widehat{R}_s(\mathbf{1}) = qf(0)g(h/s). \quad (73)$$

Squaring and summing over $s \neq 0$ proves (70). $\square$

Remark 5.2 (Additive centering does not remove the principal character). The hypotheses $\sum f = \sum g = 0$ were used throughout, yet (70) can still be positive. Ordinary additive centering therefore does not eliminate the principal multiplicative face. A CRT or physical application must control that face separately, or impose additional vanishing such as $f(0) = 0$.

5.2 Average versus maximum

Multiplicative Parseval applied directly to $a \mapsto R_s(a)$ gives

$$\boxed{\frac{1}{q-1} \sum_{\chi} \mathcal{E}_{\chi} = \sum_{a,s \in G} |R_s(a)|^2.} \quad (74)$$

The right side is exactly the centered projective second moment from TPC-33. Thus independent complete product–slope averaging controls the *average* multiplicative spectral energy, while a coherent weighted sum in (64) has norm given by the *maximum*.

Corollary 5.3 (Sharp average-to-maximum factor). *For every centered pair f, g ,*

$$\max_{\chi} \mathcal{E}_{\chi} \leq (q-1) \sum_{a,s \in G} |R_s(a)|^2. \quad (75)$$

For every $q > 2$, equality in the factor $q-1$ is possible: there are nonzero centered f, g for which exactly one $\mathcal{E}_{\chi}$ is nonzero.

Proof. The first assertion follows from nonnegativity and (74). Choose a nonprincipal character χ_0 and put

$$f(0) = 0, \quad f(D) = \overline{\chi_0(D)} \ (D \neq 0), \quad g = \mathbf{1}_{\{0\}} - \mathbf{1}_{\{1\}}. \quad (76)$$

Both functions are centered. Character orthogonality gives

$$\sum_{D \neq 0} f(D) \chi(D) = 0 \quad \text{unless } \chi = \chi_0.$$

The principal energy vanishes because $f(0) = 0$, while the bracket in (69) is $2q-1 > 0$ at χ_0 . Hence only $\mathcal{E}_{\chi_0}$ survives and the ratio of maximum to average is $q-1$. $\square$

5.3 A finite-model coefficient-specific gate

The exact formula identifies the form of a useful new input for every layer that has already been reduced to the complete, separable form (64), with $b(\ell)$ independent of j, s . On such a layer, (67) asks not for $\max_{\chi} \mathcal{E}_{\chi}$ to be small for every possible b , but for the weighted spectral mass

$$\frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \mathcal{E}_{\chi} \quad (77)$$

to satisfy the desired ledger. A sufficient route on that finite model is to show that b has little mass on characters with large $\mathcal{E}_{\chi}$, or that the Möbius-derived f, g have small $\mathcal{E}_{\chi}$ on the character support of b .

The actual multiplier $\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j)$ depends jointly on the input row, the opposite row, and j ; it also retains the d -variable, smooth weights, gcd projector, and near-row deletion. We have not proved an $X^{o(1)}$ -cost decomposition of that multiplier into complete separable product–slope layers, nor the completion of its j, s ranges. Thus (77) is an exact energy only after this missing physical interface lemma has been supplied.

No physical spectral-overlap estimate or reassembly lemma is proved here. The formula is nevertheless more specific than a uniform operator norm: it exposes the signed finite Mellin data a future theorem would have to use after reassembly. At $q \asymp J$, the sharp average–to–maximum factor $q-1 \asymp J$ coincides with the missing energy factor in the optimistic one–modulus ledger (60).

6 CRT alias compression and its non-tensor boundary

The single-modulus conflict in TPC-33 is elementary: orbit coverage prefers $q \lesssim J$, whereas no-wrap on an affine determinant of size X requires $q \gg X$. Several small moduli can increase the joint modulus without sacrificing local orbit coverage. We now compute the exact threshold and then prove why independent one-coordinate bounds do not automatically multiply.

6.1 Exact alias identity

Theorem 6.1 (Finite CRT alias decomposition). *Let $q_1, \dots, q_r$ be positive integers, let*

$$M = [q_1, \dots, q_r], \quad (78)$$

and let $F \in \mathbb{Z}$ satisfy $|F| \leq B$. Then

$$\prod_{i=1}^r \mathbf{1}_{q_i|F} = \mathbf{1}_{M|F} = \sum_{|k| \leq \lfloor B/M \rfloor} \mathbf{1}_{F=kM}. \quad (79)$$

For all integers F in the ambient interval $|F| \leq B$, the joint congruence is equivalent to $F = 0$ if and only if $M > B$.

Proof. Divisibility by every q_i is equivalent to divisibility by their least common multiple M . Every multiple $F = kM$ in the stated interval has $|k| \leq \lfloor B/M \rfloor$, proving the identity. If $M > B$, the only possible k is zero. Conversely, if $M \leq B$, the value $F = M$ is a nonzero counterexample to a uniform no-wrap assertion on the whole ambient interval. $\square$

Applied to

$$F = sU - aD - h, \quad (80)$$

whose physical ambient range is $B \asymp X$, this is an exact decomposition of a congruence relaxation into parallel integer determinant fibers. It does not bound the mass on any one fiber.

Corollary 6.2 (The orbit-scale alias ladder). *Suppose $q_1, \dots, q_r$ are pairwise coprime and $q_i \asymp J = X^{133/400+o(1)}$. Then the number of possible k in (79) is*

$$O\left(1 + \frac{X}{J^r}\right), \quad \frac{X}{J^r} = X^{1-r(133/400)+o(1)}. \quad (81)$$

In particular,

r	X/J^r	Interpretation
1	$X^{267/400+o(1)}$	one-modulus alias family
2	$X^{134/400+o(1)}$	two-modulus alias family
3	$X^{1/400+o(1)}$	first diagonal-budget scale
4	$X^{-132/400+o(1)}$	strict no-wrap for large X

Among orbit-scale choices $q_i \asymp J$, four moduli are the first that force strict no-wrap on an $O(X)$ ambient determinant interval.

Because $QJ = X^{1+o(1)}$, the three-modulus endpoint is

$$\frac{X}{J^3} = \frac{Q}{J^2} = X^{1/400+o(1)}. \quad (82)$$

This is the same factor by which the allowed orbit energy Q^3/J exceeds the proved diagonal Q^2J . Hence the following accounting statement is exact: if a joint three-modulus decomposition has $O(X^{o(1)}X/J^3)$ alias fibers and every fiber can be reassembled at scale $O(X^{o(1)}Q^2J)$, then absolute reassembly gives

$$\frac{X}{J^3} Q^2J = \frac{Q^3}{J}. \quad (83)$$

Neither the fiber estimate nor its uniformity is proved here. In particular, (82) is exponent compatibility, not closure of the physical gate.

6.2 The exact two-congruence compatibility condition

We record a small point needed to avoid a false CRT simplification. TPC-19 contains the primitive determinant–divisor normal form, while TPC-29 proves the general compatibility $g \mid h(m_1 - m_2)$, the unique lcm class, and its primitive specialization [4, 7]. The self-contained restatement below makes the reduced divisor $g/(g, h)$ explicit and records a useful counterexample; the compatibility principle itself is not new.

Proposition 6.3 (Affine CRT compatibility). *Let $u_1, u_2 \geq 1$, let $g = (u_1, u_2)$, and assume $(m_i, u_i) = 1$. Then the congruences*

$$m_i j \equiv -h \pmod{u_i}, \quad i = 1, 2, \quad (84)$$

are compatible if and only if

$$\boxed{g \mid h(m_1 - m_2)}, \quad (85)$$

equivalently

$$\frac{g}{(g, h)} \mid m_1 - m_2. \quad (86)$$

When compatible, they determine one residue class modulo $[u_1, u_2]$. Under the physical primitive hypothesis

$$(u_i, hm_i) = 1, \quad i = 1, 2, \quad (87)$$

compatibility reduces to

$$\boxed{(u_1, u_2) \mid m_1 - m_2}. \quad (88)$$

Proof. The individual solutions are

$$j \equiv -hm_i^{-1} \pmod{u_i}.$$

The generalized CRT says they are compatible exactly when the two residues agree modulo g :

$$hm_1^{-1} \equiv hm_2^{-1} \pmod{g}.$$

Multiplication by m_1m_2 , which is invertible modulo g , gives (85). Dividing by the common factor (g, h) gives (86). Under (87), $(g, h) = 1$, giving (88). The modulus of the joint class is the least common multiple by the generalized CRT. $\square$

The condition $(m_i, hu_i) = 1$ is not a substitute for (87): it makes m_i invertible but need not make h invertible modulo u_i . For example, $h = 2, u_1 = u_2 = 4, m_1 = 1, m_2 = 3$ gives compatible congruences although $4 \nmid m_1 - m_2$; the general criterion (85) is correct.

6.3 CRT lifts do not force tensor gain

Let $M = M_0 M'$, where $(M_0, M') = 1$. For $f, g : \mathbb{Z}/M_0\mathbb{Z} \rightarrow \mathbb{C}$, lift them to $\mathbb{Z}/M\mathbb{Z}$ by

$$F(D) = f(D \bmod M_0), \quad G(U) = g(U \bmod M_0). \quad (89)$$

For units $a, s \pmod{M}$, put

$$\mathcal{C}_M(a, s) := \sum_{D \bmod M} F(D) G(s^{-1}(aD + h)). \quad (90)$$

Let $\bar{F} = M^{-1} \sum_D F(D)$ and similarly for all means.

Theorem 6.4 (Proper-coordinate non-tensorization). *With a_0, s_0, h_0 denoting reduction modulo M_0 , one has*

$$\mathcal{C}_M(a, s) - M\bar{F}\bar{G} = M'(\mathcal{C}_{M_0}(a_0, s_0) - M_0\bar{f}\bar{g}). \quad (91)$$

Consequently

$$\begin{aligned} & \sum_{a, s \in (\mathbb{Z}/M\mathbb{Z})^\times} |\mathcal{C}_M(a, s) - M\bar{F}\bar{G}|^2 \\ &= M'^2 \varphi(M')^2 \sum_{a_0, s_0 \in (\mathbb{Z}/M_0\mathbb{Z})^\times} |\mathcal{C}_{M_0}(a_0, s_0) - M_0\bar{f}\bar{g}|^2. \end{aligned} \quad (92)$$

In the normalized rms

$$\text{rms}_M^2 := \frac{1}{\varphi(M)^2} \sum_{a, s \in (\mathbb{Z}/M\mathbb{Z})^\times} \left| \frac{\mathcal{C}_M(a, s) - M\bar{F}\bar{G}}{M} \right|^2, \quad (93)$$

the value is exactly $\text{rms}_{M_0}^2$.

Proof. Every residue $D_0 \pmod{M_0}$ has exactly M' lifts modulo M , while the lifted summand depends only on D_0, a_0, s_0 . Hence

$$\mathcal{C}_M(a, s) = M' \mathcal{C}_{M_0}(a_0, s_0), \quad \bar{F} = \bar{f}, \quad \bar{G} = \bar{g},$$

which proves (91). Each pair (a_0, s_0) has $\varphi(M')^2$ unit lifts. Squaring the pointwise identity and summing gives (92). Finally use $\varphi(M) = \varphi(M_0)\varphi(M')$ and $M = M_0 M'$ in (93); all M' -factors cancel. $\square$

Taking M_0 to be the product of any proper subset of the prime coordinates shows that every proper-coordinate face of a squarefree CRT product may retain the normalized rms of the smaller product. Extra coordinates only repeat data that do not depend on them. This is the orbit-scale form of the low-coordinate phenomenon already encountered in earlier conductor decompositions [5, 6]. A joint three-modulus proof must control principal, singleton, and double-coordinate modes; the full-coordinate piece alone does not suffice.

6.4 Alias-coverage bookkeeping for a merged modulus

One might merge small moduli into P , use $O(1 + X/P)$ aliases, and then complete the J -long orbit modulo P , at cost $O(1 + P/J)$. This specified route obeys an exact bookkeeping inequality.

Proposition 6.5 (Merge-then-complete cost ledger). *For every $P, X, J > 0$,*

$$\left(1 + \frac{X}{P}\right) \left(1 + \frac{P}{J}\right) \geq \frac{X}{J}. \quad (94)$$

When $X \asymp QJ$, the right side is $\asymp Q$.

Proof. Expanding the left side produces the nonnegative terms $1 + X/P + P/J$ in addition to X/J . $\square$

At $P \asymp J^3$, the alias factor reaches the critical $X/J^3 = X^{1/400+o(1)}$, but the completion factor is J^2 ; their product still contains $X/J \asymp Q$. This proposition is not a no-go theorem for a joint delta method, circle method, multi-modulus large-sieve estimate, or tensor spectral estimate. It rules out only the strategy that first forgets the separate CRT coordinates and then pays for complete coverage of the merged modulus.

7 A physical same-ultra saving after orbit slicing

The preceding sections diagnose abstract routes. We now return to the actual multiplier (21), coefficient vectors, and ultra increments. Orbit slicing removes one time sum from the shared-divisor estimate of TPC-34 and yields a genuine physical saving.

7.1 Opening the divisor copies

For the left slice (23), define

$$Z_{\gamma,j,u}^L := a_R(u) \sum_{\substack{\alpha \\ u|N_\alpha(j)}} \gamma_\alpha^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j), \quad T < u \leq U_0. \quad (95)$$

Then

$$Y_{\gamma,j}^L = \sum_u Z_{\gamma,j,u}^L. \quad (96)$$

The diagonal in the two opened ultra-divisor copies is

$$\mathcal{V}_{L,u=} := \sum_{\gamma,j} \sum_{T < u \leq U_0} |Z_{\gamma,j,u}^L|^2. \quad (97)$$

Define $Z_{\alpha,j,u}^R$ and $\mathcal{V}_{R,u=}$ by exchanging the two row systems. These quantities are nonnegative. They are not, however, lower or upper bounds for V_L, V_R , because the distinct- u cross terms in (96) may have either sign.

For a termwise absolute comparison with TPC-34, also define

$$\tilde{\mathcal{V}}_{L,u=} := \sum_{\gamma,j} \sum_{T < u \leq U_0} |a_R(u)|^2 \left(\sum_{\substack{\alpha \\ u|N_\alpha(j)}} |\gamma_\alpha^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)| \right)^2, \quad (98)$$

and symmetrically $\tilde{\mathcal{V}}_{R,u=}$. The triangle inequality gives

$$\mathcal{V}_{L,u=} \leq \tilde{\mathcal{V}}_{L,u=}, \quad \mathcal{V}_{R,u=} \leq \tilde{\mathcal{V}}_{R,u=}. \quad (99)$$

7.2 Residue occupancy at one time

Lemma 7.1 (Fixed- (u, j) row occupancy). *On the physical support,*

$$\#\{\alpha : u \mid N_\alpha(j)\} \ll_{\mathcal{D}} 1 + \frac{Q}{u} \leq 1 + \frac{Q}{T}. \quad (100)$$

Proof. Primitivity (20) gives $(u, j) = 1$. Hence

$$u \mid m_\alpha j + h \iff m_\alpha \equiv -hj^{-1} \pmod{u}.$$

The distinct row labels m_α lie in an interval of length $O_{\mathcal{D}}(Q)$, so one residue class modulo u contains $O_{\mathcal{D}}(1 + Q/u)$ of them. Since $u > T$, the second bound follows. $\square$

Theorem 7.2 (T -saving bound for the opened shared-ultra block). *On the physical packet,*

$$\boxed{\mathcal{V}_{L,u=} + \mathcal{V}_{R,u=} \leq \tilde{\mathcal{V}}_{L,u=} + \tilde{\mathcal{V}}_{R,u=} \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon Q^2 J \left(1 + \frac{Q}{T}\right).} \quad (101)$$

At the endpoint (34),

$$\boxed{\mathcal{V}_{L,u=} + \mathcal{V}_{R,u=} \leq \tilde{\mathcal{V}}_{L,u=} + \tilde{\mathcal{V}}_{R,u=} \ll X^{o(1)} \frac{Q^3 J}{T}.} \quad (102)$$

Proof. For fixed γ, j, u , Cauchy–Schwarz and lemma 7.1 give

$$\begin{aligned} & |a_R(u)|^2 \left(\sum_{\substack{\alpha \\ u|N_\alpha(j)}} |\gamma_\alpha^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)| \right)^2 \\ & \leq |a_R(u)|^2 \#\{\alpha : u \mid N_\alpha(j)\} \sum_{\substack{\alpha \\ u|N_\alpha(j)}} |\gamma_\alpha^{(1)}|^2 |\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)|^2 \\ & \ll X^\varepsilon \left(1 + \frac{Q}{T}\right) \sum_{\substack{\alpha \\ u|N_\alpha(j)}} |\gamma_\alpha^{(1)}|^2. \end{aligned}$$

Here logarithmic factors in a_R and the bounded multiplier are absorbed into X^ε . Sum first over u . For each fixed (α, j) , the number of divisors of $N_\alpha(j) \asymp X$ is $O_\varepsilon(X^\varepsilon)$. Then sum over $O(Q)$ opposite rows γ , over $O(J)$ orbit times, and use (17). This yields

$$\tilde{\mathcal{V}}_{L,u=} \ll X^\varepsilon \left(1 + \frac{Q}{T}\right) QJ \cdot Q.$$

The right side is identical. Combine with (99). Since $Q/T \rightarrow \infty$ at the endpoint, (102) follows. $\square$

The absolute orbit–sliced baseline is $Q^3 J$, so (102) saves the full factor T . It is also one factor J below the two–time shared-ultra scale $Q^3 J^2/T$ proved in TPC-34. Nevertheless

$$\frac{Q^3 J/T}{Q^3/J} = \frac{J^2}{T} = X^{279/1000+o(1)}, \quad (103)$$

so the block does not close the orbit gate.

7.3 Terminal atoms on the same- u diagonal

On the endpoint packet, the terminal divisor

$$u = N_\alpha(j) = m_\alpha j + h \quad (104)$$

lies in the ultra range. At one fixed orbit time, equal terminal divisors have no off–diagonal row solutions.

Proposition 7.3 (Terminal equality is the row diagonal). *If $j > 0$ and*

$$N_{\alpha_1}(j) = N_{\alpha_2}(j), \quad (105)$$

then $m_{\alpha_1} = m_{\alpha_2}$, hence $\alpha_1 = \alpha_2$ under the row-label convention.

Proof. Subtracting the two equalities gives $(m_{\alpha_1} - m_{\alpha_2})j = 0$. Since $j > 0$, the row labels agree. $\square$

Thus the terminal–terminal contribution inside the same- u block is already contained in the proved Gram–copy diagonal. This does not remove the terminal obstruction identified in TPC-34: two distinct terminal divisors occur in the cross- u part of $|\sum_u Z_{\gamma,j,u}^L|^2$, where the fused signs produce an averaged four–Möbius row correlation. Nor does the triangle inequality between terminal and proper-divisor layers give a converse bound, because those layers may cancel before squaring. The theorem here proves a T -saving bound for one opened positive block and locates the remaining signed interaction more precisely; it does not break the parity boundary.

8 Short-orbit CRT and ambient integer incidence

Large ultra divisors give strong uniqueness along a J -long orbit. This section separates that fact from a different question: how many distinct divisor atoms can land at the same orbit time or integer point.

8.1 An exact short-orbit incidence norm

Let $I \subset \mathbb{Z}$ be an interval of diameter strictly smaller than every modulus u_x in a finite atom set $\mathcal{X}$. Associate to atom x one residue class $r_x \pmod{u_x}$, and define the incidence matrix

$$B_{x,j} = \mathbf{1}_{j \equiv r_x \pmod{u_x}}, \quad x \in \mathcal{X}, \quad j \in I. \quad (106)$$

Proposition 8.1 (Exact short-orbit norm). *Every row of B has at most one nonzero entry, and*

$$\|B\|_{\text{op}}^2 = \max_{j \in I} \#\{x \in \mathcal{X} : j \equiv r_x \pmod{u_x}\}. \quad (107)$$

Proof. Two distinct integers in I differ by less than u_x , so they cannot belong to the same residue class modulo u_x . Thus every row contains at most one 1. It follows that B^*B is diagonal, and its j -th diagonal entry is the number of atoms incident to j . Its largest eigenvalue is the right side of (107). $\square$

For a physical atom (m, u) , primitivity gives $(u, hm) = 1$ and the residue is $r \equiv -hm^{-1} \pmod{u}$. Since $u > T > J$ at the endpoint, the proposition applies to every ultra atom. For two atoms, [proposition 6.3](#) determines whether their residue classes have a joint return modulo the least common multiple. The remaining difficulty is precisely the column degree in (107): uniqueness for each atom does not bound how many different atoms choose the same time.

8.2 An integer fixed-determinant transform

Let $h \neq 0$ be fixed. Consider primitive coefficient pairs $\lambda = (a, s)$ in a box where the a -interval has length at most A and $1 \leq s \leq S$. Associate the integer line

$$L_{a,s} = \{(x, y) \in \mathbb{Z}^2 : sy - ax = h\}, \quad (a, s) = 1. \quad (108)$$

Let $\mathcal{P} \subset \mathbb{Z}_{>0}^2$ be contained in a box whose x -interval has length at most X_0 and whose points satisfy $y > T$. Define

$$(\mathcal{R}F)(a, s) = \sum_{P \in L_{a,s} \cap \mathcal{P}} F(P). \quad (109)$$

For a point P , let

$$r(P) = \#\{(a, s) : P \in L_{a,s}\}. \quad (110)$$

Lemma 8.2 (Primitive-line Schur bound). *One has*

$$\|\mathcal{R}\|_{\text{op}}^2 \leq \max_{(a,s)} \sum_{P \in L_{a,s} \cap \mathcal{P}} r(P). \quad (111)$$

Proof. The kernel of $\mathcal{R}\mathcal{R}^*$ is the nonnegative intersection count

$$(\mathcal{R}\mathcal{R}^*)((a,s), (a',s')) = |L_{a,s} \cap L_{a',s'} \cap \mathcal{P}|.$$

Its row sum is $\sum_{P \in L_{a,s} \cap \mathcal{P}} r(P)$. Schur's test proves the result. $\square$

Two distinct primitive lines have at most one rational intersection. If $a's - as' \neq 0$, it is

$$x = \frac{h(s' - s)}{a's - as'}, \quad y = \frac{h(a' - a)}{a's - as'}. \quad (112)$$

The next theorem gives the corresponding integral degree bound without discarding parallel or nonintegral cases.

Theorem 8.3 (Ambient integer-incidence norm). *With the preceding hypotheses, every point $P = (x, y) \in \mathcal{P}$ satisfies*

$$r(x, y) \ll_h 1 + \min \left\{ \frac{A}{y}, \frac{S}{x} \right\} \leq_h 1 + \frac{A}{T}, \quad (113)$$

and every line $L_{a,s}$ in the coefficient family satisfies

$$|L_{a,s} \cap \mathcal{P}| \leq 1 + \frac{X_0}{s}. \quad (114)$$

Consequently

$$\|\mathcal{R}\|_{\text{op}}^2 \ll_h \left(1 + \frac{X_0}{s_{\min}} \right) \left(1 + \frac{A}{T} \right), \quad (115)$$

where $s_{\min}$ is the least s in the coefficient family.

Proof. Fix $P = (x, y)$. Any two coefficient solutions of $sy - ax = h$ differ by

$$\Delta s = \frac{x}{(x, y)} k, \quad \Delta a = \frac{y}{(x, y)} k, \quad k \in \mathbb{Z}. \quad (116)$$

If P lies on one of the lines, then $(x, y) \mid h$, so $(x, y) \leq |h|$. Counting the admissible k 's in the a - and s -intervals gives the first estimate in (113); $y > T$ gives the second.

Now fix a primitive line. Its integer points satisfy

$$ax \equiv -h \pmod{s}.$$

Because $(a, s) = 1$, this is one residue class for $x \pmod{s}$. An interval of length X_0 contains at most $1 + X_0/s$ such integers, and y is then unique. This proves (114). Insert the two bounds into lemma 8.2 and maximize over the line family. $\square$

8.3 Physical containing-box comparison

On a projected affine layer from TPC-33, the coefficient and point boxes have scales

$$A \asymp LJv, \quad X_0 \asymp D/v, \quad 1 \leq v \leq D, \quad (117)$$

and the reflected complementary slope has $s \geq 1$. Hence

$$\begin{aligned} \|\mathcal{R}\|_{\text{op}}^2 &\ll_h \left(1 + \frac{D}{s_{\min}v}\right) \left(1 + \frac{LJv}{T}\right) \\ &\ll_h 1 + \frac{X}{T} \end{aligned} \quad (118)$$

at the endpoint. Indeed $LJD = QJ \asymp X$, the cross term is at most X/T , and $D \leq X/T$ there.

This containing-box bound is sharp in scale. Take $v \asymp D$, a point $x = 1, y \asymp T$, and coefficients

$$a = sy - h. \quad (119)$$

For $\gg_h X/T$ values of $s \leq cX/T$ coprime to h , these are primitive lines in an a -box of length $O(X)$ passing through the same point. Thus one can have $r(P) \gg_h X/T$ in the ambient box.

After extracting the outer row L^2 mass Q , the uniform orbit-Gram ledger would require an unsigned incidence constant of scale Q/J . The ratio supplied by the containing-box relaxation is

$$\frac{X/T}{Q/J} = \frac{J^2}{T} = X^{279/1000+o(1)}, \quad (120)$$

the same gap as (103). This does not prove that the physical incidence transform is sharp: the actual point set is selected by squarefree and Möbius weights, gcd projectors, smooth supports, and coupled row labels. It says that the unsigned ambient box alone reproduces, rather than removes, the shared-ultra loss. No norm comparison between the actual masked orbit Gram matrix and this containing-box transform is asserted.

9 Endpoint ledger and the next coefficient-specific gate

The results above have different logical status. The Gram lower bounds hold for abstract phase classes, the projective saturation theorem concerns a particular right-hand side, the character formula is an exact finite diagonalization, the CRT ladder is exact arithmetic bookkeeping, and the same-ultra estimate is a physical upper bound. We compare them without turning a method limitation into a physical lower bound.

9.1 Energy ledger

At

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)},$$

the orbit-energy scales are as follows.

Table 1: Orbit-sliced energy ledger at the high- β endpoint.

Object	Scale	X -exponent
Absolute orbit-sliced baseline	$Q^3 J$	4670/2000
One-modulus multiplicity certificate, optimistically applied	Q^3	4005/2000
Physical same-ultra block	$Q^3 J/T$	3898/2000
Sufficient orbit-sliced gate	Q^3/J	3340/2000
Proved orbit-sliced diagonal	$Q^2 J$	3335/2000
Three-modulus aliases times diagonal	$(X/J^3)Q^2 J$	3340/2000

The last row is conditional accounting: it assumes a diagonal-scale bound for every alias fiber. The equality of its exponent with the gate is the critical match (82), not a proved estimate.

The uniform norm comparison is even more decisive within the abstract phase class.

Table 2: Uniform Gram scales in the equal-deletion phase model.

Quantity	Abstract lower	Required scale	Ratio
Operator norm	Q^2	Q^2/J	J
Hilbert–Schmidt norm squared	$Q^4 J$	Q^4/J^2	J^3

These ratios explain why replacing the physical coefficient direction by all unit vectors is too expensive in a broad model. They do not determine the norms of the actual masked matrices.

9.2 Two viable next formulations

The exact product spectrum suggests a signed finite-field target. On a layer already reduced to the complete separable form (64), the quantity to control is

$$\mathfrak{E}_q(b; f, g) := \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \mathcal{E}_{\chi}, \quad (121)$$

By theorem 5.1, this is exactly the coherent energy of that finite layer. To feed it into (33), one must first decompose the actual j -dependent mask and incomplete ranges into such layers with total cost $X^{o(1)}$. That physical interface lemma is not proved here. After it is supplied, a theorem showing that the resulting $\widehat{b}$'s avoid characters with large $\mathcal{E}_{\chi}$ would be sufficient; it need not bound $\max_{\chi} \mathcal{E}_{\chi}$ for every abstract coefficient vector.

The CRT formulation keeps three pairwise coprime primes $q_1, q_2, q_3 \asymp J$ separate. Its character expansion has eight coordinate faces:

$$\emptyset, \quad \{1\}, \{2\}, \{3\}, \quad \{1, 2\}, \{1, 3\}, \{2, 3\}, \quad \{1, 2, 3\}. \quad (122)$$

The empty face is the principal term; singleton and double faces are the non-tensor modes exposed by theorem 6.4; the full face is where three independent character oscillations may multiply. A genuine joint theorem must separately control or eliminate the principal face; ordinary additive centering does not do so automatically, as (70) shows. The proper faces must likewise be small, absorbed into previously bounded diagonals, or cancelled by the actual coefficients. Estimating only the full face is insufficient.

These are two descriptions of the same strategic requirement: do not replace the actual coefficient direction by a maximum over all characters, and do not replace joint CRT sampling by three independent applications of a one-modulus mean square.

9.3 Route menu

The next step may proceed along any of the following nonexclusive routes.

Route A: *Coefficient-weighted multiplicative spectrum.* First prove a controlled complete separable decomposition of the physical mask, then establish a dyadically averaged bound for (121), retaining the Möbius row coefficients, prime-source restriction, and actual gcd projector. It is enough to control spectral overlap, not every character separately.

Route B: *Joint three-modulus dispersion.* Expand the three CRT coordinates simultaneously, center before separating them, and prove diagonal-scale reassembly for all aliases

and proper-coordinate modes. The exact available alias budget is $X/J^3 = Q/J^2$; four moduli may be used for no-wrap, but they add a further coordinate face that also requires control.

Route C: *Terminal/proper signed coupling*. Use the terminal equality result to remove the same-divisor terminal off-diagonal, disperse the proper-divisor layers through their complementary factors, and retain cancellation between terminal and proper layers. A triangle inequality separating them may destroy the needed cancellation.

Route D: *Hybrid incidence-spectrum*. Use short-orbit uniqueness and gcd/lcm pruning to lower the multiplicity of the bad character fibers, then apply the coefficient-specific spectrum only to the remaining high-degree points. The ambient Schur bound (118) shows that incidence without signed spectral input is insufficient.

The first two routes are now precise enough that failure can also be made mathematical. A construction concentrating the physical coefficient mass on one bad character, or a lower-coordinate CRT mode surviving all available centerings, would be a scoped negative result. Conversely, a positive estimate for only a restricted range of projector divisors or alias fibers would still close a nontrivial subpacket.

9.4 Relation to the preceding chain

TPC-19 supplies primitive determinant-divisor CRT normal forms [4]. TPC-29 already proves the general reflected compatibility $g \mid h(m_1 - m_2)$, its primitive specialization, and the unique lcm class [7]. The self-contained restatement in proposition 6.3 adds the reduced criterion $g/(g, h) \mid m_1 - m_2$ and records a counterexample to a tempting swapped coprimality hypothesis; the compatibility theorem itself is not claimed as new. TPC-22 and TPC-24 already show that low conductor coordinates can survive a product decomposition [5, 6]; the new use here is at orbit-scale projective moduli with the explicit three/four threshold. TPC-33 proves the unweighted complete product-slope lift and fixed-determinant projective Plancherel identity [9]. The new spectral step is theorem 5.1, which diagonalizes a coherent weighted product-slope sum. TPC-34 first states the coefficient-specific orbit gate, proves its diagonal, and gives a coherent constant-phase witness at the sign-blind scale [10]. The present rank-trace theorem upgrades that witness to a lower bound for every unit-phase equal-deletion frame, with Fourier sharpness. The new physical step is theorem 7.2, which proves a T -saving bound for the opened same-ultra block after orbit slicing.

Even a future proof of (33) would close only the inherited matched residual shell. Earlier packet reconstruction, main-term identification, and positivity stages would remain. The word “dynamical” here refers to deterministic affine traversal $j \mapsto mj + h$ modulo divisor moduli. No mixing theorem, logistic-map conjugacy, Möbius disjointness theorem, or spectral interpretation of zeta zeros is asserted.

10 Exact certificate and claim boundary

The accompanying standard-library certificate is

`experiments/tpc35_certificate.py.`

It uses integers, `Fraction`, and hand-written Gaussian-rational arithmetic. It contains no floating-point calculation, random input, true division, or Python `assert` statement. Running

```
python experiments/tpc35_certificate.py
python -O experiments/tpc35_certificate.py
```

performs the same checks and regenerates byte-identical JSON.

10.1 Finite checks

The archived run performs 141,803 exact checks.

Table 3: Exact finite checks in the TPC-35 certificate.

Check family	Count
Aspect-ratio Gram, Hilbert–Schmidt, and tight-frame identities	1,260
Same-source block identity and Rayleigh lower bound	1,311
General and primitive ultra-CRT compatibility and occupancy	95,535
Short-orbit congruence operator	393
Incomplete multiplicity saturation	18,414
Product-slope character spectrum over $q = 3, 5$	194
Three/four-modulus alias ladder	8,736
CRT lift and normalized non-tensorization	1,338
Integer-line intersection, Schur count, and sharp pencil	14,618
Source constraints	4
Total	141,803

The Gram suite constructs the matrices M_j entry by entry, forms $\sum_j M_j^* M_j$, and compares it to (42). It checks the complete eigenvalue-free operator and Hilbert–Schmidt inequalities and exact tight-frame sharpness using integer Hadamard phases. The same-source suite checks (52) and the rt , rather than r^2t , subtraction in (53) through an exact Rayleigh witness.

The CRT suite enumerates the general criterion (85), the primitive specialization (88), the least-common-multiple joint class, and exact interval occupancy. It includes the explicit $h = 2, u_1 = u_2 = 4, m_1 = 1, m_2 = 3$ counterexample to the incorrectly swapped coprimality hypothesis. The short-orbit suite compares the direct singular norm with the maximum time degree in (107).

The projective suite checks the three pigeonhole ranges and the continuous envelope algebra with rational arithmetic. The product-slope suite uses Gaussian rationals to verify every transform, energy, principal and nonprincipal formula for all centered test functions in finite exact families over $q = 3, 5$. It also verifies the single-character example attaining the factor $q - 1$. The alias suite checks exact floor counts for the lcm fibers and the rational exponent ledger. The CRT-lift suite compares both pointwise centered correlations and normalized rms values. The integer-line suite verifies intersection formulas, solution steps, point and line multiplicities, Schur row sums, and a sharp common-point pencil.

10.2 Reproducibility values

The archived outputs have:

$$\begin{aligned} \text{normalized source SHA-256} \quad & 7e08d1d29dc88e40b9653176b06336ba \\ & 2ed879db33fb080e2c697e79b94e948e, \end{aligned} \quad (123)$$

$$\begin{aligned} \text{JSON SHA-256} \quad & 02dea583a12f7078ee15ae8205f7e8d0 \\ & 787f92c5c858c80a525396534357c134, \end{aligned} \quad (124)$$

$$\begin{aligned} \text{certificate digest} \quad & 26b282b0b8a989e792524dc12ab1a6ed \\ & 8215204fc36a0ef4493ab28982a7dae5. \end{aligned} \quad (125)$$

Line endings are normalized before the source hash is taken. The certificate digest hashes the canonical JSON payload before its digest field is appended.

10.3 What is proved and what remains open

The unconditional abstract results are the equal-deletion and same-source Gram identities and lower bounds, the saturation of the residue-multiplicity certificate, the exact product-slope character spectrum, the CRT alias identity and threshold, proper-coordinate non-tensorization, the merge-complete cost ledger, the short-orbit operator identity, and the ambient integer-incidence bound. The unconditional physical result is the same-ultra estimate (101), together with terminal equality on that block.

Finite enumeration certifies algebra, finite incidence, and rational bookkeeping. It does not prove:

- (i) asymptotic divisor, coefficient, or smooth-weight envelopes;
- (ii) a lower bound for either physical orbit-Gram norm;
- (iii) that the physical uniform operator gate is false;
- (iv) an upper or lower bound for the actual correlation from the multiplicity saturation theorem;
- (v) suppression of bad multiplicative characters for the physical coefficient vector;
- (vi) a complete separable product-slope decomposition of the actual orbit-sliced mask with total cost $X^{o(1)}$;
- (vii) a joint three-modulus dispersion or diagonal-scale alias-fiber estimate;
- (viii) control of the distinct-ultra cross terms;
- (ix) cancellation in the terminal four-Möbius row correlation;
- (x) the coefficient-specific orbit gate (33);
- (xi) the inherited Q^3 column-energy or scalar-shell gate;
- (xii) an affine Chowla theorem, Möbius disjointness theorem, or breach of sieve parity;
- (xiii) positivity or a Hardy-Littlewood prime-pair asymptotic; or
- (xiv) infinitely many twin primes.

The paper therefore advances the proof program in a deliberately bounded way: two tempting uniform shortcuts are shown to saturate in their stated abstract classes, one coherent product-slope operator is exactly diagonalized, the first orbit-scale alias depth compatible with the diagonal budget is identified, and one physical divisor block is improved. The next unproved input is coefficient-sensitive cancellation in (121), or a joint CRT theorem strong enough to imply it after physical reassembly. This scoped boundary is consistent with the classical parity phenomenon in sieve theory [2]; it is not a parity-breaking theorem.

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